

Real-time Dictionary based Super-Resolution of Surveillance Video Streams and Targets

Timothy M. Hospedales and Shaogang Gong
Vision Semantics Ltd, UK.

Keywords: Visual surveillance, super-resolution, sparse coding

ABSTRACT

Real-time super-resolution within surveillance video streams is a powerful tool for security and crime prevention allowing for example, events, faces or objects such number-plates and luggage to be more accurately identified on the fly and from a distance. However, many of the state of the art approaches to super-resolution are computationally too expensive to be suitable for real-time applications within a surveillance context. We consider one particular contemporary method based on sparse coding,¹ and show how, by relaxing some model constraints, it can be sped up significantly compared to the reference implementation, and thus approach real-time performance with visually indistinct reduction in fidelity. The final computation is three orders of magnitude faster than the reference implementation. The quality of the output is maintained: PSNR of the super-resolved images compared to ground truth is not significantly different to the reference implementation, while maintaining a noticeable improvement over baseline bicubic-interpolation approach.

1. INTRODUCTION

The effectiveness of visual surveillance systems in practice is often limited by the resolution of the capture device relative to the area of the space being surveyed. Persons, objects and events observed at a distance from the camera are captured with few pixels, often providing insufficient detail for identification. Enhancement of image detail via super-resolution would provide a powerful capability for visual surveillance. It could enhance ability of the operator to, for example, identify people, determine if an object or event is suspicious or read a license plate.

Super-resolution has been extensively studied in the computer graphics and vision community for offline enhancement of image detail – see references for classic and recent examples^{2,3} and for a survey.⁴ Various ancillary directions for image enhancement have also been studied including combining multiple video frames worth of data in super-resolution,^{5,6} building super-resolution models for specific objects such as faces,^{7,8} and reversing noise processes such as blur⁹ in the data. However, these are complementary to the core single image super-resolution task considered here.

In this paper we develop a significant speedup to a state of the art sparse coding (SC) super-resolution algorithm.^{1,8,10,11} The principle of sparse coding using over-complete dictionaries has emerged in recent years as a powerful approach for many image processing tasks including super-resolution,^{1,8,10,11} de-noising¹² and restoration.¹³ However, many such of the mentioned methods require solving multiple expensive constrained optimization problems to process each incoming image, or even to process every overlapping patch of each image. This thereby limits their applicability to real-time surveillance applications.

We start by reviewing a general formulation of the super-resolution problem, the specific class of solutions entailed by sparse coding methods, and our proposed improvements in Section 2. In Section 3 we evaluate the computational complexity and super-resolution fidelity of the resulting system. Finally, in Section 4, we close with some comments about how state of the art super-resolution methods can be most effectively applied in visual surveillance systems.

2. METHODS

2.1 Super-Resolution Background

Super-resolution is the problem of inverting a downsampling image formation model. An image formation model describes the observation of a low resolution image $\mathbf{y}$ given a high resolution image $\mathbf{x}$ obtained by some downsampling process:

$$\mathbf{y} = SH\mathbf{x} + \epsilon, \quad (1)$$

where matrix $H : R^h \rightarrow R^h$ is a blur filter, matrix $S : R^h \rightarrow R^l$ is a downsampling matrix and ϵ represents Gaussian noise. Since the dimension of $\mathbf{y} \in R^l$ after downsampling is much less than $\mathbf{x} \in R^h$ ($h \gg l$), directly inverting this process to obtain $\mathbf{x}$ from $\mathbf{y}$ is under-constrained. That is, there are many (even an infinite number of) high resolution images $\mathbf{x}$ which correspond to a single low-resolution image $\mathbf{y}$ under the formation model Eq. (1), so a direct unique solution is impossible. Many approaches therefore leverage a probabilistic representation of the super-resolution process:

$$p(\mathbf{x}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{x})p(\mathbf{x}). \quad (2)$$

In this case, statistics from real-world data can be leveraged to alleviate the under-constrained nature of the problem: While an infinite number of potential high resolution images may correspond to each low resolution image, we can search for the high resolution image which matches the low resolution image *and is also likely to occur based on real-life image statistics*. This knowledge is learned from training data D , either in the form of (i) pairs $\{\mathbf{x}_i, \mathbf{y}_i\}$ of high and low resolution images, from which a direct mapping $p(\mathbf{x}|\mathbf{y})$ can be learned; or (ii) high resolution images $\{\mathbf{x}_i\}$ images alone, from which the prior statistics $p(\mathbf{x})$ of images can be learned to constrain Eq. (2) and permit a MAP solution. In both cases, super-resolution of $\mathbf{x}$ from $\mathbf{y}$ is then posed as searching for the *most likely* $\mathbf{x}$ given $\mathbf{y}$ and the learned prior knowledge:

$$\mathbf{x}^* = \underset{\mathbf{x}}{\operatorname{argmax}} p(\mathbf{x}|\mathbf{y}, D). \quad (3)$$

2.2 Sparse Coding

The sparse coding model of^{1,10,11} learns a direct mapping from low to high resolution via pairs $D = \{\mathbf{x}_i, \mathbf{y}_i\}$ of high and low resolution training images (i.e., directly modeling Eq. (3)). Specifically, a large over-complete set of *corresponding* high and low resolution patches D_h and D_l are sampled from $\{\mathbf{x}_i\}$ and $\{\mathbf{y}_i\}$ respectively. These patch sets are referred to as *dictionaries*. The assumption of this model is that the training set D is representative of real-world images, and hence each patch in any new image can be represented by a sparse combination of patches from the dictionary. Since the high and low resolution dictionaries D_h and D_l correspond, such a *sparse encoding* of a low resolution image $\mathbf{y}$ automatically induces the most likely corresponding high resolution image $\mathbf{x}$. Specifically, under this approach, the image formation process is approximated as:

$$\begin{aligned} \mathbf{y} &= SH\mathbf{x} + \epsilon, \\ &\approx SHD_h\boldsymbol{\alpha} + \epsilon, \end{aligned} \quad (4)$$

where the high resolution image $\mathbf{x}$ is encoded in terms of a sparse representation $\boldsymbol{\alpha}$, $\|\boldsymbol{\alpha}\|_0 \ll h$ and a dictionary D_h . In this framework, super-resolution is then posed as the optimization problem of finding a sparse encoding of the low-resolution image:

$$\boldsymbol{\alpha}^* = \arg \min_{\boldsymbol{\alpha}} \|\mathbf{D}_l \boldsymbol{\alpha} - \mathbf{y}\|_2^2 + \lambda \|\boldsymbol{\alpha}\|_0, \quad (5)$$

where the parameter λ trades-off the objective of sparsity (L0-norm of $\boldsymbol{\alpha}$) and the match of the encoding to the observed low resolution image $\mathbf{y}$. Given $\boldsymbol{\alpha}^*$, the most likely high-resolution image can be constructed directly via a sparse combination of elements from the corresponding high resolution dictionary:

$$\mathbf{x}^* = \mathbf{D}_h \boldsymbol{\alpha}^*. \quad (6)$$

The process in Eq. (5)-(6) is performed for each individual patch. To complete the super-resolution process for the entire image, additional constraints may be imposed on the overall image $\mathbf{x}$ to ensure that adjacent patches are chosen to be smooth, or every overlapping patch may be processed and the results averaged to obtain the final high resolution output $\mathbf{x}^*$.^{1,10,11} Additionally, constraints may also be applied to ensure that the resulting reconstruction matches the low-resolution image under the formation model^{1,10} (i.e., Eq. 4 may also be enforced).

2.3 Speedup

For sparse coding based super-resolution, we discovered a good compromise between fidelity and efficiency of online processing by relaxing the sparsity constraint during reconstruction. The dictionaries $\{\mathbf{D}_l, \mathbf{D}_h\}$ are still trained by standard over-complete dictionary learning techniques.¹ However, relaxing the sparsity constraint at reconstruction time allows the code $\boldsymbol{\alpha}$ to be computed directly based on L2-norm optimization of the coded dictionary elements and the low resolution input image $\mathbf{y}$. From Eq. (5) we obtain:

$$\boldsymbol{\alpha}^* = \arg \min_{\boldsymbol{\alpha}} \|\mathbf{D}_l \boldsymbol{\alpha} - \mathbf{y}\|_2^2, \quad (7)$$

$$\approx \mathbf{D}_l^+ \mathbf{y}, \quad (8)$$

$$\mathbf{x}^* = \mathbf{D}_h \boldsymbol{\alpha}^*, \quad (9)$$

$$= \mathbf{D}_h \mathbf{D}_l^+ \mathbf{y}, \quad (10)$$

where $\mathbf{D}_l^+$ is the Moore-Penrose pseudo-inverse of the low resolution dictionary. Since $\mathbf{D}_l$ is over-complete and with more columns than rows, this procedure finds the encoding $\boldsymbol{\alpha}^*$ for which $\mathbf{D}_l \boldsymbol{\alpha} = \mathbf{y}$ (of which there are many) and $\|\boldsymbol{\alpha}\|_2^2$ is minimal (of which there is one). That is, the low-resolution image is encoded in terms of the dictionary, and then the high resolution image is reconstructed using this encoding. This approximation provides an order of magnitude speedup compared to the reference implementation¹ of Eq. (5), which needs to perform iterative optimization to obtain a sparse (low L0-norm) code.

In contrast to the inefficient iteration over patches in,¹ the proposed simplification of the per-patch super-resolution procedure to vector operations also permits all the image patches to be stacked into matrix $\mathbf{Y}$ and hence the whole image $\mathbf{X}$ to be efficiently super-resolved in four large matrix operations:

$$\begin{aligned} \mathbf{A}^* &\approx \mathbf{D}_l^+ \mathbf{Y}, \\ \mathbf{X}^* &= \text{unstack}(\mathbf{D}_h \mathbf{A}^*), \end{aligned} \quad (11)$$

where *unstack* denotes summing out corresponding pixels in overlapping patches. This means that the procedure is simple to implement, because it requires only four calls to linear-algebra libraries. Moreover it is fast to compute due, requiring solely linear-algebra libraries which are available in highly optimized form for any given hardware platform. In practice this provides another order of magnitude speedup. Finally, by simplifying the whole image procedure into a single set of matrix operations, Eq. (11) is also amenable to efficient execution on a graphics processing unit (GPU) implementation unlike Eqs. (5)-(6),¹ which provides a final order of magnitude speedup.

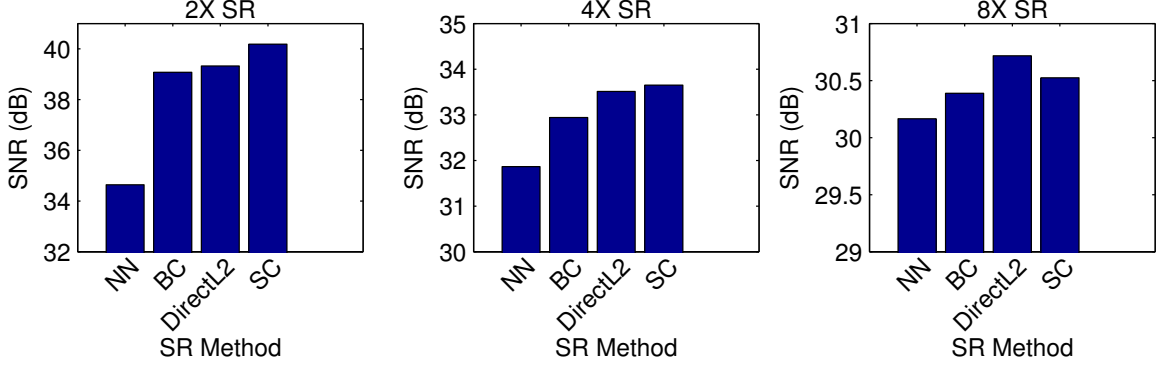


Figure 1: SNR Comparison for our DirectL2 versus baselines.

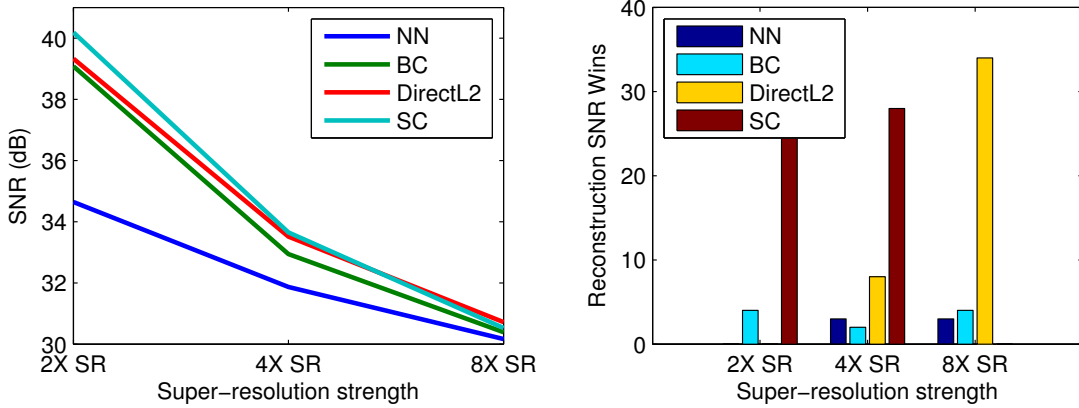


Figure 2: SNR Comparison of our DirectL2 versus baselines. Left: Super-resolution SNR at each magnification. Right: Number of wins for each model out of 40 images.

3. EXPERIMENTS

We compare our new model (denoted DirectL2) against two three baselines: a naive nearest-neighbor upscaling (denoted NN), the industry standard approach of bicubic interpolation (denoted BC), which should be considered lower baselines; as well as an upper baseline of the reference implementation of sparse coding super-resolution¹ (denoted SC). To evaluate the methods on realistic surveillance data, we collected 40 face images from a variety of surveillance data sources including the benchmark UK home office iLIDS and iLIDS MCT datasets¹⁴ as well as datasets collected by ourselves from the london underground, and by BAA from an operationally deployed system at Heathrow airport. We evaluate 2X, 4X and 8X super-resolution conditions – corresponding to increasingly small objects and an increasingly challenging super-resolution scenario. The original images ranged from 64-256 pixels in dimension, and we down-sampled them 2-8X before super-resolving them again to attempt to reconstruct the original resolution.

3.1 Comparison of Super-Resolution Fidelity

In the first experiment, we compute the mean SNR between the ground truth image and the super-resolved image as shown in Figs. 1 and 2. In the simplest case of 2X super-resolution DirectL2 performs only slightly better than the BC lower baseline, and not as well as the SC upper baseline. However, in the more realistic case (a human operator is more likely to need assistance to identify a target if it is smaller to start with) of 4X super-resolution, DirectL2 performs about as well as the SC upper baseline. In the most challenging case of 8X super-resolution, DirectL2 actually outperforms the reference SC implementation.

Typical examples of super-resolution results at each scale are shown in Fig. 3. In most cases, both sparse coding methods (columns 3 and 4) are significantly sharper than the baseline bicubic method (column 2).

Meanwhile, the Fast SC method (column 3) is usually at least as sharp as the reference SC method (column 4). Most interestingly, the raw NN images at the lowest input resolution (top row of each set) would be impossible to identify, while some of the 8X super-resolved images would arguably be identifiable. Ground truth images of each person are shown at the bottom of Fig. 3.

3.2 Comparison of Super-Resolution Computation Cost

To compare the speed of our DirectL2 method and the reference SC implementation,¹ we recorded the computation time required to super-resolve the batch of surveillance images from the previous section with 2X, 4X and 8X magnification factors. Fig. 3.2 summarizes the computation time per image as a function of the size of the output image on a log-log scale. Color indicates the reference SC method versus DirectL2, while symbols indicate the degree of magnification.. As expected, the computation cost for each model is linear in the size of the output image. So a larger initial image with smaller magnification and a smaller initial image with larger magnification can take similar time (overlapping symbols).

Our DirectL2 method implemented in matlab, is about 2 orders of magnitude or more faster than the reference implementation throughout the range of tasks*. Crucially, for up to about 10^4 output pixels, DirectL2 approaches real time performance (10 frames per second). Exploiting matlab GPU computation to execute Eq. (11), DirectL2-GPU obtains almost a further order of magnitude improvement in computation speed over DirectL2. Overall, DirectL2-GPU provides real time performance for up to about 10^5 output pixels. This corresponds to about a 25x25 pixel image at 4X magnification, or a 12x12 pixel at 8X magnification. We did not execute the largest images at 8X resolution on the GPU because at this point GPU memory limitations prevent storing all the patches in memory simultaneously, and a more complicated implementation would be necessary to overcome this.

3.3 Practical Application Considerations

Application Scenarios Although we have demonstrated how to achieve dramatic speed increase over the reference implementation¹ of SC super-resolution; in practice even with an optimized implementation, real-time super-resolution of an entire video stream would require prohibitive computation in the near future. Nevertheless, one of the most typical operator tasks is to identify all persons in a scene, possibly against a prior watch-list.¹⁵ In this section, we briefly discuss some application scenarios in which the improved super-resolution capabilities of state of the art SC algorithms could be exploited more selectively to assist in this task would requiring full-frame real-time performance.

Faces: To avoid costly super-resolution of the entire video stream, while supporting a typical operator task of identifying persons, a standard fast person¹⁶ or face¹⁷ detector could be applied to the video stream, and the super-resolution algorithm applied solely to detected body or face regions. When applied solely to face regions, the computation requirements are within the real-time capabilities of even our matlab implementation.

Window: Many surveilled spaces involve gauntlet situations, such as doorways or corridors which every person entering or exiting the scene must pass through. An appropriately defined super-resolution sub-window therefore provides the opportunity to enhance at least some images of all person entering or exiting a scene without requiring full-frame processing. This is illustrated in Fig. 5 for the iLIDS¹⁴ underground and doorway monitoring datasets.

Gallery: Performing live super-resolution on video streams or windows thereof may not be the most effective use of the technology. As people walk through a scene, their identifiability is affected by head direction, shadows and motion blur, etc. Indeed a person may only be identifiable for only a small subset of frames within a video, which could easily be missed by an operator monitoring the live stream. An alternative approach is to apply tracking and association to maintain a list of unique identities of people in the scene,¹⁵ and then present in a separate window a gallery of the most identifiable image(s) so far of each unique person in the scene. Identifiability is ranked by (i) positive frontal face detection,¹⁷ and (ii) high score

*Both codes are implemented in matlab

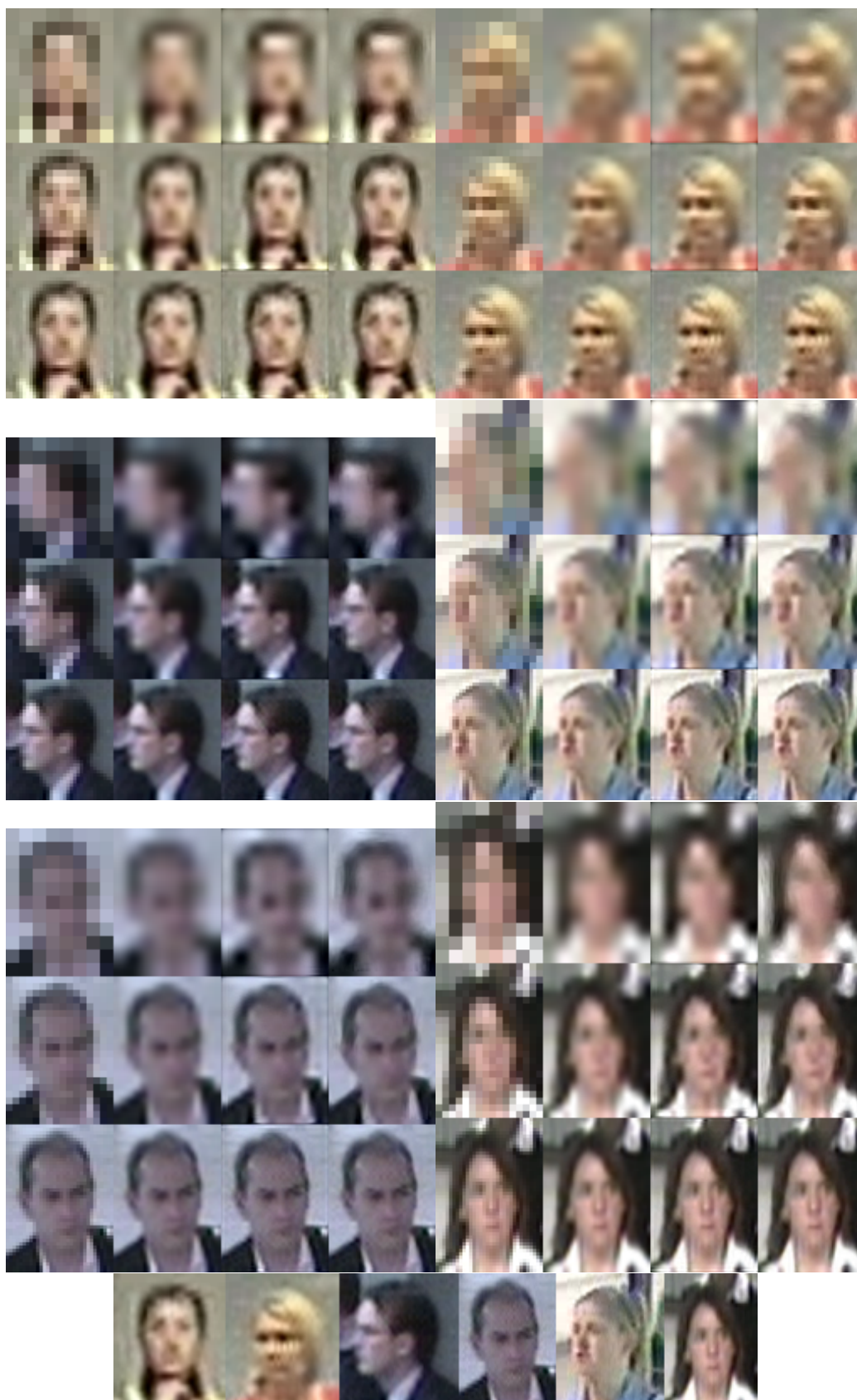


Figure 3: Super-resolution examples at 2X, 4X and 8X (rows) for NN, BC, DirectL2 and SC methods (cols).

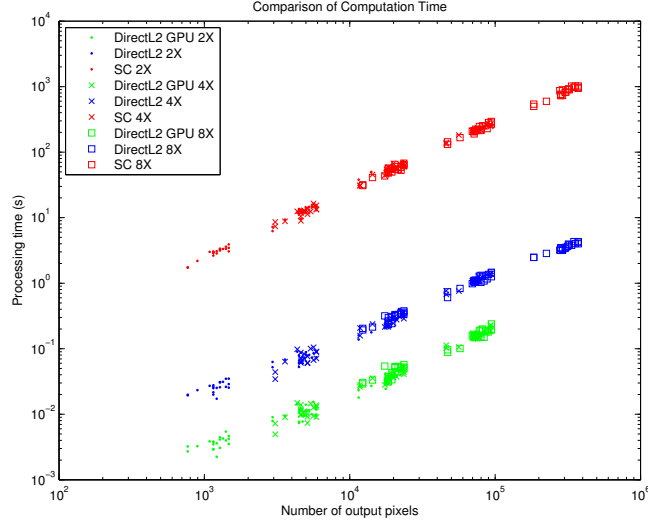


Figure 4: Super-resolution computation time for a collection of images. Color indicates method SC versus DirectL2, while symbols indicate the degree of magnification.



(a) Inpainted super-resolution ($4\times$) window in a corridor. (iLIDS underground dataset)
(b) Super-resolution ($4\times$) window for a doorway (iLIDS doorway monitoring dataset). Right: NN vs DirectL2 super-resolution.

Figure 5: Example applications of a super-resolution window.

on a blind image quality assessment metric.¹⁸ Finally, those images presented in the gallery (the most identifiable image so far) needs to be super-resolved. The operator can conveniently monitor this small gallery view to match against his/her watch list.

Image Quality Finally, we briefly mention the impact of image quality (as opposed to resolution per se) for super-resolution. We note that the super-resolution image-formation model Eq. (1) takes no account of technological noise such as compression artifacts and interlacing which are common in surveillance video. In our experiments, we observed that SR algorithms typically respond very to this type of noise. For practical use, video streams should ideally be de-interlaced in advanced if necessary, and only lightly compressed with a modern codec.

4. CONCLUSIONS

In this paper, we have analyzed a state of the art super-resolution method¹ based on sparse coding. Although this method provides significantly improved super-resolution fidelity over industry standard bicubic interpolation, it does so at massive computational cost, prohibiting its use in real-time applications such as surveillance. We discover a novel method and demonstrate its unique advantages for breaking the necessary computational barriers demanded by real-world applications. In particular, our method relaxes assumptions of the current benchmark sparse coding super-resolution method whilst retaining fidelity. It allows a significantly faster realization of the algorithm, as well as maintaining the a significant margin of improvement in SNR over the bicubic baseline – especially at high magnification factors, which are exactly the situations where super-resolution would be necessary to help a surveillance operator identify a distant and/or small object. Finally, we discussed some practical application scenarios where super-resolution methods can be applied to good effect even if super-resolving a full-frame video stream is too costly for real-time computation.

REFERENCES

- [1] Yang, J., Wright, J., Huang, T. S., and Ma, Y., “Image super-resolution via sparse representation,” *IEEE Transactions on Image Processing* **19**, 2861–2873 (2010).
- [2] Freeman, W. T., Pasztor, E. C., and Carmichael, O. T., “Learning low-level vision,” *International Journal of Computer Vision* **40**, 25–47 (Oct. 2000).
- [3] He, H. and Siu, W.-C., “Single image super-resolution using gaussian process regression,” in [*IEEE Conference on Computer Vision and Pattern Recognition*], IEEE Computer Society (2011).
- [4] van Ouwerkerk, J., “Image super-resolution survey,” *Image and Vision Computing* **24**(10), 1039 – 1052 (2006).
- [5] Irani, M. and Peleg, S., “Improving resolution by image registration,” *CVGIP: Graph. Models Image Process.* **53**, 231–239 (Apr. 1991).
- [6] Liu, C. and Sun, D., “A bayesian approach to adaptive video super resolution,” in [*IEEE Conference on Computer Vision and Pattern Recognition*], IEEE Computer Society (2011).
- [7] Jia, K. and Gong, S., “Generalized face super-resolution,” *IEEE Transactions on Image Processing* **17**(6), 873–886 (2008).
- [8] Yang, J., Tang, H., Ma, Y., and Huang, T. S., “Face hallucination via sparse coding,” 1264–1267 (2008).
- [9] Levin, A., Weiss, Y., Durand, F., and Freeman, W. T., “Efficient marginal likelihood optimization in blind deconvolution,” in [*IEEE Conference on Computer Vision and Pattern Recognition*], 2657–2664 (2011).
- [10] Yang, J., Wright, J., Huang, T., and Ma, Y., “Image super-resolution as sparse representation of raw image patches,” in [*IEEE Conference on Computer Vision and Pattern Recognition*], 1–8 (2008).
- [11] Zeyde, R., Elad, M., and Protter, M., “On single image scale-up using sparse-representations,” in [*Proceedings of the 7th international conference on Curves and Surfaces*], 711–730, Springer-Verlag (2012).
- [12] Elad, M. and Aharon, M., “Image denoising via sparse and redundant representations over learned dictionaries,” *IEEE Transactions on Image Processing* **15**(12), 3736–3745 (2006).
- [13] Zoran, D. and Weiss, Y., “From learning models of natural image patches to whole image restoration,” in [*IEEE International Conference on Computer Vision*], 479–486 (2011).
- [14] HOSDB, “Imagery library for intelligent detection systems (i-lids),” in [*IEEE Conf. on Crime and Security*], (2006).
- [15] Raja, Y., Gong, S., and Xiang, T., “User-assisted visual search and tracking across distributed multi-camera networks,” in [*Proc. SPIE 8189, 81890Q. Security and Defence Conference on Optics and Photonics for Counter-terrorism and Crime Fighting.*], (2011).
- [16] Felzenszwalb, P. F., Girshick, R. B., McAllester, D., and Ramanan, D., “Object detection with discriminatively trained part-based models,” *IEEE Transactions on Pattern Analysis and Machine Intelligence* **32**, 1627–1645 (2010).
- [17] Viola, P. and Jones, M. J., “Robust real-time face detection,” *International Journal of Computer Vision* **57**, 137–154 (2004).
- [18] Ferzli, R. and Karam, L. J., “A no-reference objective image sharpness metric based on the notion of just noticeable blur (jnb),” *IEEE Transactions on Image Processing* **18**, 717–728 (April 2009).